CS105 **Spr'26**

Mini - Project (Lab 5 – 6)

Due: submission – May 8 (Needs to be uploaded to Canvas and Gradescope),
demo by May 13.
Late submission and demo – May 15 (20% off)!

Your assignment is to design and conduct a survey, and then analyze and present the data collected from the students of CS105 and CS111 classes. The dataset will be available in Google Drive.

Decide on a topic of your study and create a survey that will allow you to collect the appropriate data (as discussed in class). (If you don't submit your questions by the deadline, you can use the available dataset, but you will lose 20% of your score.)

When the master dataset becomes available, download it and create your own dataset. (Remove all columns that you won't need, clean the data that you plan to analyze.) Now perform exploratory data analysis (EDA). Use Jupyter notebooks.

Project Tasks and Guiding Questions
Your project should address the following questions and steps, using **Jupyter notebooks**:

1. **What data do you have?**
   Describe the dataset **you** are using.
2. **What would you like to know?**
   Clearly state the questions or problems you aim to investigate.
3. **Explore and analyze the data.**
   Compute relevant statistics and create visualizations to explore patterns and relationships in the data.
4. **State hypotheses or predictions and test them.**
   Formulate hypotheses and identify appropriate statistical tests to evaluate them.
   Apply the selected tests and interpret the results.
5. **Overall conclusion.**
   Summarize your findings and discuss what they suggest in relation to your original questions.

Project Deliverables
Your submission should include the following components:

- **Topic overview**
  Introduce the topic and provide context for the data and research questions.
- **Questions 1 and 2**
  Clearly describe the data you are using and state what you aim to learn from it.

- **Exploratory Data Analysis (EDA):**

  I. Data Exploration and Visualization
  Create at least 8 visualizations, with at least 6 different types.
  These visualizations should go beyond basic pie or bar charts and combine variables in ways that reveal relationships, patterns, or insights from the data.
    - For each visualization:
      1. Explain what you are trying to show or learn with this visual.
      2. Perform the necessary computations and generate the visualization using Python.
      3. Analyze and interpret the results in the context of your question or hypothesis.

  II. Formulating and Testing Hypotheses
  Clearly state at least 3 hypotheses or predictions based on your EDA.
    - For each hypothesis:
      1. Describe and justify the statistical test you are using.
      2. Perform the test using Python.
      3. Analyze the results and draw a conclusion.
    - Use at least 2 different statistical tests across all hypotheses: at least one correlation analysis and a chi-squared test.

- **Overall conclusion**
  Summarize your findings and relate them back to the original questions.

- **Group contributions**
  Describe each group member's contribution to the project.

Submission instructions:

1. The first submission for your project topic and questions (via Google drive) is **due by Thursday, April 16 .** If revisions are necessary, you may edit the topic and/or the questions and resubmit. The final version must be ready by Monday**, April 20, at 6pm.** After this deadline, we will begin preparing the master survey, and any further changes may not be included.) You may be asked to meet with the instructor or one of the graders to clarify your questions (the notifications will be sent over Slack).

2. Once you completed the project, you need to submit it to canvas (ipnb) and Gradescope (pdf) by the deadline, one per group.

To submit to Gradescope:
Go to File > Export Notebook As >PDF.
Double check that the entire notebook is in this PDF file. If not, try first exporting the notebook to HTML, and then printing to PDF.
Upload the PDF to Gradescope.

3. Demo: You will need to demo your project in class or during the available demo time. All group members must be present during the demo.